

UNIT 1 · PART 7 · SLIDES 54–61

Introduction to Agentic AI

From simple rule-based programs to autonomous systems that **think, plan, and act** — understanding the next frontier of artificial intelligence.





SLIDE 54

Why Do We Need Agentic AI?

Imagine asking an AI assistant: *"Plan my 4-day Goa trip under ₹30,000, book flights, reserve a hotel, check the weather, and prepare an itinerary."*

Classical AI Limitations

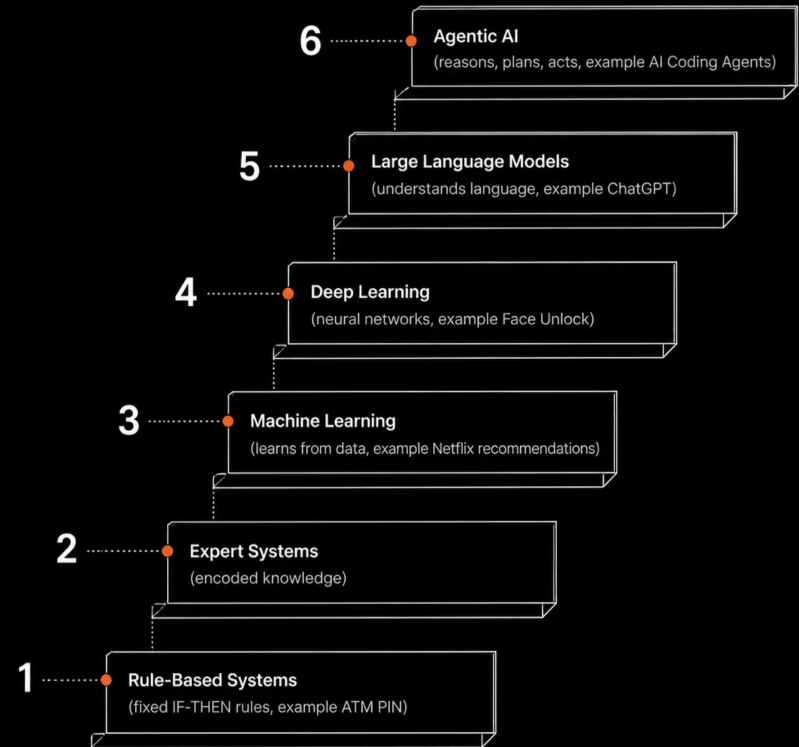
- Designed for single, specific tasks
- No multi-step planning or memory
- Cannot access external tools autonomously
- Require continuous human guidance

What Modern AI Must Do

- Understand user goals in natural language
- Plan and execute multiple connected steps
- Search the web, read documents, use APIs
- Learn from feedback and complete tasks autonomously

The Evolution of Artificial Intelligence

AI has evolved from rigid rule-based programs to autonomous systems capable of reasoning, planning, and independent action. Each generation built upon the last, growing more flexible and powerful.



What is Agentic AI?

Agentic AI is an AI system that understands goals, reasons about problems, plans multiple steps, uses external tools, and executes tasks **autonomously** — behaving like a skilled digital employee rather than a simple chatbot.

Goal-Oriented

Works toward a defined objective

Reasoning

Analyzes problems before acting

Multi-Step Planning

Breaks tasks into sub-steps

Tool Usage

Calls APIs, searches, databases

Memory

Retains context across interactions

Adaptability

Learns from feedback and adjusts

Traditional AI vs. Agentic AI

Dimension	Traditional AI	Agentic AI
Scope	Performs one task	Completes entire workflows
Behavior	Reactive	Proactive
Planning	No planning	Multi-step planning
Memory	No persistent memory	Uses memory across sessions
Reasoning	Limited	Advanced
Tools	Cannot use external tools	Uses APIs, search, databases
Autonomy	Requires frequent user input	Works autonomously toward goals

 **Key Takeaway:** Traditional AI solves a single problem. Agentic AI completes an entire workflow — from goal to result.

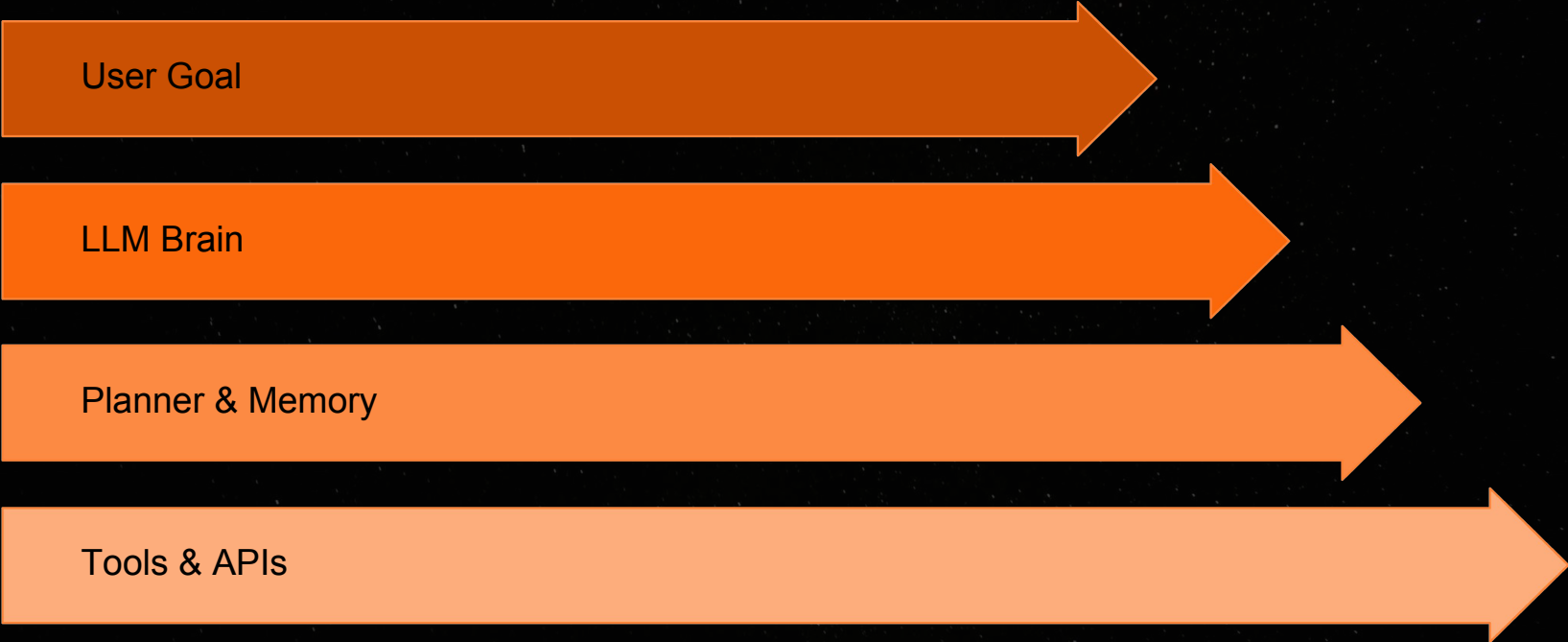
Characteristics of Agentic AI



- **Autonomy**
Operates without constant human oversight
- **Planning & Reasoning**
Decomposes complex goals into actionable steps
- **Tool Integration**
Connects to search engines, APIs, and databases
- **Memory & Feedback**
Remembers context and improves from outcomes

Components of an Agentic AI System

An Agentic AI system is not just an LLM — it is an **ecosystem** of reasoning, planning, memory, and tools working together.



User Goal

LLM Brain

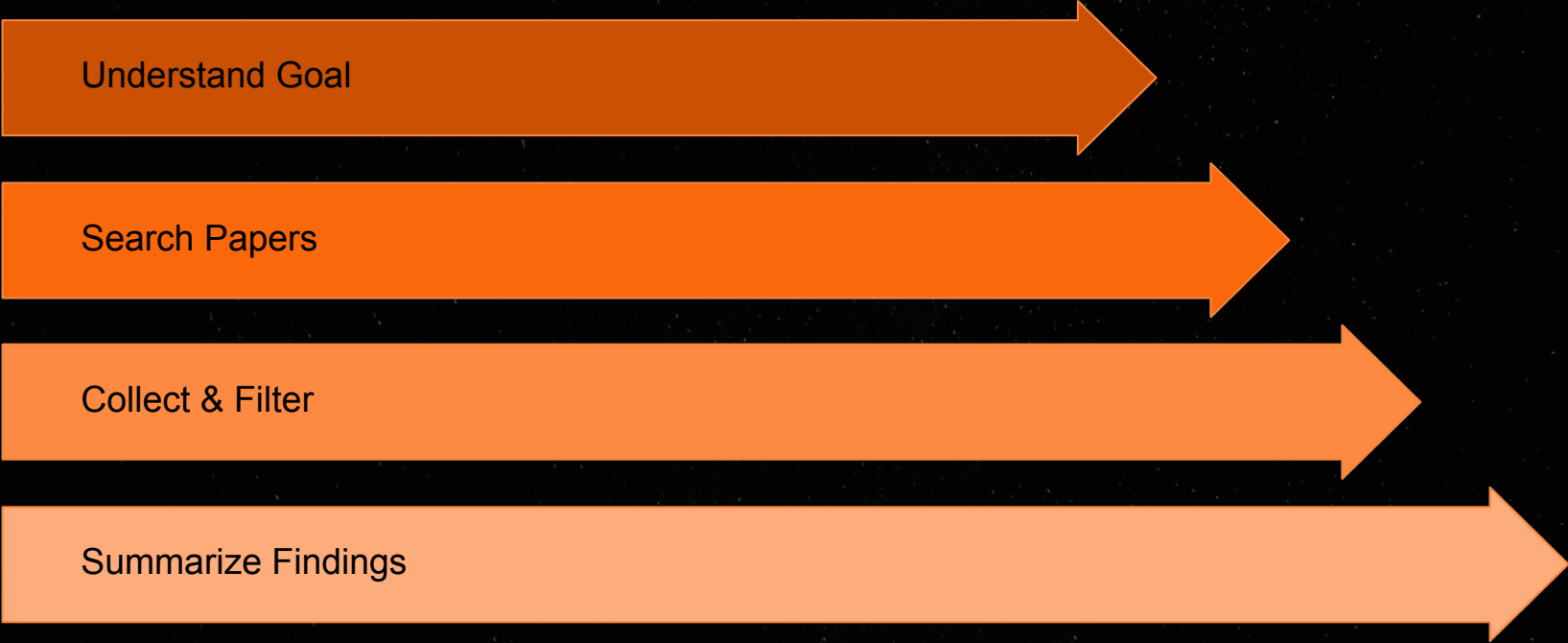
Planner & Memory

Tools & APIs

The **LLM** acts as the brain, the **Planner** breaks tasks into steps, **Memory** retains context, **Tools** perform external actions, and the **Feedback Loop** refines future decisions.

Case Study: AI Research Assistant

"Prepare a report on Electric Vehicles in India."



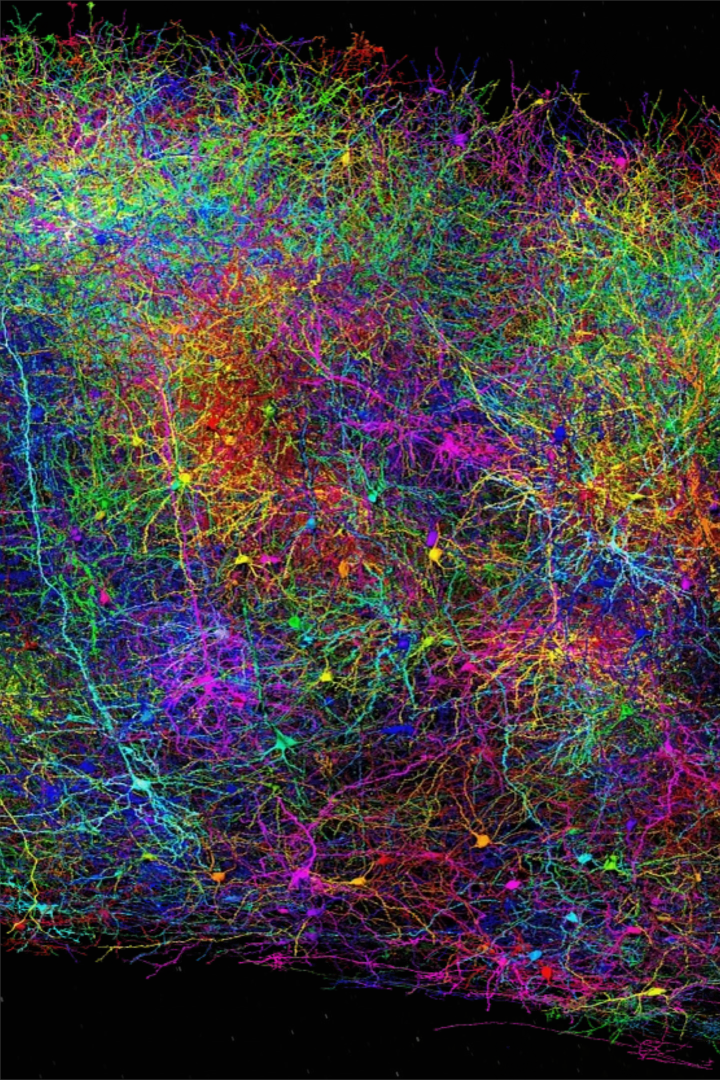
Understand Goal

Search Papers

Collect & Filter

Summarize Findings

The agent autonomously searches academic databases, filters irrelevant information, summarizes key findings, and produces a structured professional report — saving time and reducing manual effort.



SLIDE 61

Bridge to LLM-Powered Agents

We've seen that Agentic AI can understand goals, plan tasks, use tools, and make decisions. But what gives these agents the ability to **understand and reason about human language**?

The Answer

Large Language Models (LLMs) — the cognitive engine powering modern AI agents

Next Lecture Preview

What is an LLM? How LLMs become the brain of an AI agent. Agent design patterns. Real-world applications.

Key Takeaways

1

Classical AI Has Limits

Single-task, reactive systems cannot handle complex real-world workflows

2

Agentic AI is Autonomous

It reasons, plans, uses tools, and completes entire workflows independently

3

Architecture Matters

LLM + Planner + Memory + Tools + Feedback = a complete agentic system

4

LLMs Are the Brain

Next lecture explores how Large Language Models power intelligent agents